

#HW 8

● Graded

34 Minutes Late

Student

Bartosz Bielicz

Total Points

95 / 100 pts

Question 1

(no title)

30 / 30 pts

✓ - 0 pts Correct

Question 2

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 4 pts $\Delta M = \Delta t (M_{in} - M_{out})$

- 2 pts the calculation was asked for 1 hour, not 30 mins.

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 3 pts $m_1 = m_2 + m_3$

- 3 pts $m_1 = 10.27 \text{ kg/s}$, $m_2 = m_3 = 5.13 \text{ kg/s}$

- 5 pts $1 \text{ bar} = 10^5 \text{ Pa}$; $m_1 = m_2 + m_3$ where, $m_2 = m_3$.

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 3 pts $Q = m [C_p \Delta T + (V_2^2 - V_1^2)/2]$

- 5 pts rate of heat transfer not calculated/ not included in scan

- 4 pts $v_1 = 6.25 \text{ m/s}$; $v_2 = 7.37 \text{ m/s}$; $Q = 10.56 \text{ kW}$

- 2 pts $v_2 = 7.37 \text{ m/s}$

Question 5

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 5 pts $v_2 = \sim 149 \text{ ft/s}$; check the unit conversions carefully.

- 7 pts incomplete execution of solution

Question 6

(no title)

10 / 15 pts

- 0 pts Correct

- 3 pts 133.1 kW

✓ - 5 pts 133.1 kW; check the calculation clearly.

Questions assigned to the following page: [2](#), [3](#), and [4](#)

Barthosz Bielicz

Prof. Bilgin

Homework 8

CHE 201

2. $\dot{m}_1 = 0.8 \text{ lb/s}$ $\dot{m}_2 = 1.2 \text{ lb/s}$ $\dot{m}_3 = 2.5 \text{ lb/s}$ $\dot{m}_i = 2000 \text{ lb/s}$

$1 \text{ h} \cdot 3600 \text{ s} = 3600 \text{ s}$

$\dot{m}_{\text{net}} = (0.8 + 1.2) - 2.5 = -0.5 \text{ lb/s}$

$-0.5 \text{ lb/s} \cdot 3600 \text{ s} = -1800 \text{ lb}$

$2000 \text{ lb} - 1800 \text{ lb} = \boxed{200 \text{ lb left}}$

3. S.S. $\rightarrow \dot{m}_1 = \dot{m}_2 + \dot{m}_3$

$v_1 = \frac{RT_1}{P_1} = \frac{(0.4815)(350 + 273)}{(5 \text{ bar} \cdot 10^2 \text{ kPa})} = 0.584 \text{ m}^3/\text{kg}$

$\dot{m}_1 = \frac{\dot{V}A}{v} = \frac{(30 \text{ m}^3/\text{s})(0.2 \text{ m}^2)}{(0.584 \text{ m}^3/\text{kg})} = 10.27 \text{ kg/s}$

$\dot{m}_2 = \dot{m}_3 \rightarrow \frac{10.27 \text{ kg/s}}{2} = 5.135 \text{ kg/s}$

$\boxed{\dot{m}_1 = 10.27 \text{ kg/s} \quad \dot{m}_2 = 5.135 \text{ kg/s} \quad \dot{m}_3 = 5.135 \text{ kg/s}}$

4. $T_1 = 290 \text{ K}$ $P_1 = 1 \text{ bar}$ $\dot{V} = 0.25 \text{ m}^3/\text{s}$, $T_2 = 325 \text{ K}$ $P_2 = 0.95 \text{ bar}$ $A = 0.04 \text{ m}^2$

a) $\dot{m} = \frac{\dot{V}}{v}$ $v = \frac{RT}{P} = \frac{(0.287 \text{ kJ/kg} \cdot \text{K})(290 \text{ K})}{(1 \text{ bar} \cdot 10^2 \text{ kPa})} = 0.8323 \text{ m}^3/\text{kg}$

$\dot{m} = \frac{0.25 \text{ m}^3/\text{s}}{0.8323 \text{ m}^3/\text{kg}} = \boxed{0.3 \text{ kg/s}}$

b) $\dot{V} = vA \rightarrow v = \frac{\dot{V}}{A} \rightarrow v_1 = \frac{0.25 \text{ m}^3/\text{s}}{0.04 \text{ m}^2} = \boxed{6.25 \text{ m/s}}$

$v_2 = \frac{RT}{P} = \frac{(0.287)(325 \text{ K})}{(0.95 \text{ bar} \cdot 10^2 \text{ kPa})} = 0.982 \text{ m}^3/\text{kg}$

$\dot{V}_2 = \dot{m}v = (0.3 \text{ kg/s})(0.982 \text{ m}^3/\text{kg}) = 0.295 \text{ m}^3/\text{s}$

$$V_2 = \frac{\dot{V}_2}{A} = \frac{(0.295 \text{ m}^3/\text{s})}{0.01 \text{ m}^2} = \boxed{7.375 \text{ m/s}}$$

$$c) \quad K=1.4, \quad C_p = \frac{KR}{K-1} \rightarrow \frac{(1.4)(0.287)}{1.4-1} = 1.0045 \text{ kJ/kg} \cdot \text{K}$$

$$\begin{aligned} \dot{Q} &= \dot{m} \left(C_p (T_2 - T_1) + \frac{V_2^2 - V_1^2}{2} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} \right) \\ &= 0.3 \left(1.0045 (325 - 240) + \frac{7.375^2 - 6.25^2}{2} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} \right) \\ &= \boxed{10.56 \text{ kW}} \end{aligned}$$

$$5. \quad T_1 = 389^\circ\text{R} \quad T_2 = 450^\circ\text{R}$$

$$0 = \dot{Q} - \dot{W} + \dot{m}_i \left(h_i + \frac{V_i^2}{2g_c} + gz_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2g_c} + gz_e \right)$$

$$\dot{m}_i = \dot{m}_e = \dot{m}$$

$$h_i + \frac{V_i^2}{2g_c} = h_e + \frac{V_e^2}{2g_c}$$

$$h = C_p T$$

$$\rightarrow C_p T_1 + \frac{V_i^2}{2g_c} = C_p T_2 + \frac{V_e^2}{2g_c}$$

$$C_p = 0.24 \text{ Btu/lbm} \cdot \text{R}$$

$$\rightarrow V_e^2 = V_i^2 + 2g_c C_p (T_1 - T_2)$$

$$g_c = 32.17 \frac{\text{lbm} \cdot \text{ft}}{\text{lbf} \cdot \text{s}^2}$$

$$\rightarrow 0.24 \text{ Btu/lbm} \cdot \text{R} \cdot 778.17 = 186.76 \frac{\text{ft} \cdot \text{lbf}}{\text{lbm} \cdot \text{R}}$$

$$V_2 = \sqrt{(869)^2 + 2(32.17)(0.24)(389 - 450)} = \boxed{148.92 \text{ ft/s}}$$

$$6. \quad \dot{m}_i = \dot{m}_e = 10 \text{ kg/min} \quad h_i = 3100 \text{ kJ/kg} \quad V_i = 30 \text{ m/s} \quad z_i = 3 \text{ m} \quad z_e = 0 \text{ m}$$

$$h_e = 2800 \text{ kJ/kg} \quad V_e = 45 \text{ m/s} \quad \dot{Q} = 1.1 \text{ kJ/kg} \cdot 10 \text{ kg/min} = 11 \text{ kJ/min}$$

$$0 = \dot{Q} - \dot{W} + \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

$$\dot{W} = 11 + 10 \left(3100 + \frac{30^2}{2} + (9.81)(3) \right) - (10) \left(2800 + \frac{45^2}{2} + (9.81)(0) \right)$$

$$\dot{W} = 2680.3 \text{ kJ/min} \cdot \frac{1 \text{ min}}{60 \text{ s}} = \boxed{44.67 \text{ kJ/s}}$$